Paleomagnetism of Some Carboniferous Glacial Varves from Central Africa

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Samples have been collected at five sites from Dwyka (Carboniferous) glacial varves in Rhodesia, Zambia, and southern Tanzania. The samples were, in general, weakly magnetized, and after thermal cleaning the directions form close groups at each site although a number of samples became too weak to be measured and had to be discarded. On restoring the strata to the horizontal at each site, the directions become steeply magnetized and a significant improvement in the grouping of the site poles is noted. The pole position for these varves is calculated to lie in South Africa at 26½°S, 26½°E with $A_{95} = 10½°$. Permo-Carboniferous poles from Africa and Australia are shown to be consistent with King's reconstruction of Gondwanaland, and they confirm the upper Paleozoic polar movement from west to east across Gondwanaland. This agreement is, however, only possible if the Dwyka series is assigned to the Lower rather than Upper Carboniferous and if some of the K3 beds in the Karroo succession in southern Tanzania are Upper Carboniferous rather than Lower Permian.

INTRODUCTION

The great Carboniferous glaciation in Southern Africa has been well documented [*du Toit*, 1921, 1954; *Flint*, 1961; *King*, 1962]. The base of the Karroo system is formed by the thick Dwyka glacial conglomerate or tillite. In the southern Cape Province, the tillite attains a thickness of 800 meters, but, north of latitude 27°S, the Dwyka series becomes insignificant and erratically developed. It appears that southern Africa was the center of a great continental ice sheet at this time.

Conglomerate beds do not provide suitable material for paleomagnetic studies, but, on the periphery of the continental ice sheet, one might expect to find occurrences of glacial varves, which are eminently suitable for paleomagnetism. In South Africa the occurrence of varved sediments in the Dwyka series is rare, although in Australia varved shales are ubiquitous during the same period [*David*, 1950]. To the north of South Africa, in Rhodesia, Zambia, and southern Tanzania, however, there are isolated occurrences of varved sediments, which are the subject of the present investigation.

GEOLOGY

The Dwyka series has usually been regarded as Upper Carboniferous in age, the fossil evidence being well summarized by *du Toit* [1954]. Chief among the fossils found in the Dwyka series are *Mesosaurus* and marine fossils, including *Eurydesma* and paleoniscid fish, which assist in correlation with India and Australia.

Varved sediments occur at the base of the Karroo succession in three areas north of the Republic of South Africa (Figure 1). In the Galula coalfield of southern Tanzania (A in Figure 1), they form the major part of the K1 beds, consisting of a basal conglomerate about 3 meters thick followed by 30 meters of varved siltstones [*Spence*, 1954]. The thickness of the varves varies from 1.5 to 25 mm, averaging about 4 per cm, so that the order of 10,000 years is represented in the sequence. No fossils have been found in the K1 beds of the Galula coal field, but these beds are overlain unconformably by the K2 coal measures, which are regarded as being Lower Ecca in age determined from an abundant *Glossopteris* flora. *Opdyke* [1964] has previously reported paleomagnetic

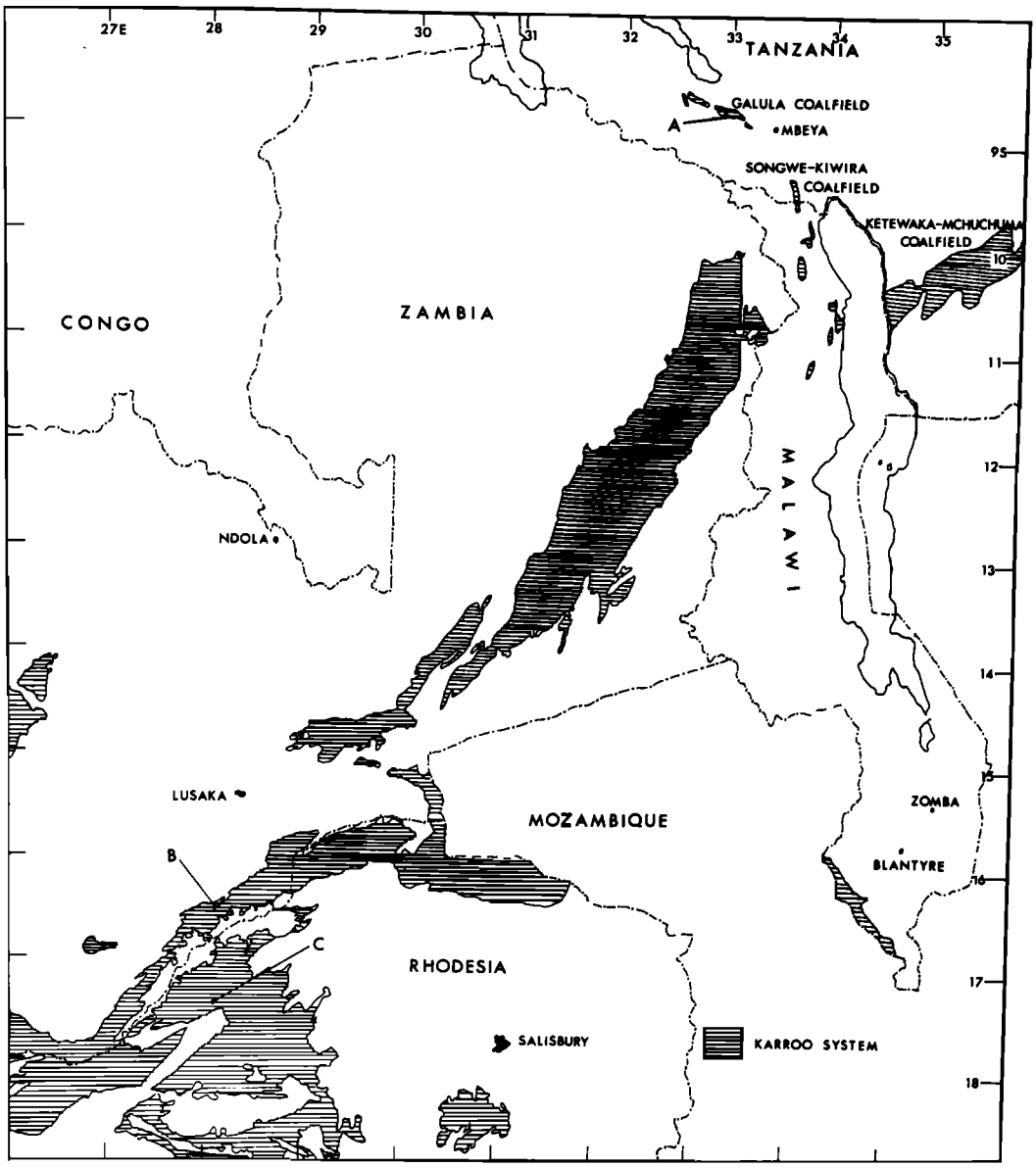


Fig. 1. Map showing the extent of the Karroo system in Rhodesia, Zambia, Malawi, and Southern Tanzania. Sampling localities of the glacial varves are indicated.

measurements from red horizons in the K3 beds of the Galula coal field.

In southern Zambia, varved clays occur in the Manga district (B in Figure 1), overlying the basal conglomerate of the Karroo system [*Gair*, 1959]. At least 12 meters of varves are present, each averaging about 10 mm in thickness. The basal beds are overlain by the Gwembe coal formation from which a species of *Glossopteris* has been indentified.

For many years evidence of the Dwyka glaciation has been lacking in Rhodesia and here the lower Wankie sandstones, regarded as being Lower Ecca in age, were designated the K1 beds. *Bond* [1952] has discovered, however, about 30 meters of tillite at the base of the

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Karroo system in the eastern Sebungwe district, and these are now designated the K0 beds. Interbedded with the tillites in the Chifuchezia River (C in Figure 1) are about 0.7-meter thickness of varved clays. The annual layers are extremely thin, the total thickness representing up to 1000 years. *Nairn* [1960] has previously reported on the NRM of four samples collected from this locality.

Sampling and Measurement

During 1962, one of us (N.D.O.) collected samples from two horizons in the K1 beds of the Galula coal field. The samples were drilled in the field with the portable coring equipment described by *McElhinny and Neal* [1967]. Each core was oriented with respect to the sun's azimuth and the horizontal by the method described by *Gough and Opdyke* [1963]. At site A1, ten cores were obtained, and at site A2, which was lower in the succession, twenty-four were obtained. During 1963 a collection of hand samples was obtained from two horizons in the varved clays from the Manga district of Zambia. At site B1, seven blocks, oriented with a Brunton compass-clinometer, were obtained, and lower in the succession at site B2 six blocks were collected. Five oriented blocks were also collected from the small occurrence of varves in Rhodesia at site C1.

The 2.5-cm-diameter cores from sites A1 and A2 were each sliced into 3.5-cm lengths. Normally, these cores are sliced into 2.5-cm length, but the samples were all very weakly magnetized and a special sample holder was used on the *Gough* [1964] spinner magnetometer to take cores of extra length to increase the magnetic moments of the specimens being measured. Although all ten samples from site A1 could be measured, only nine of the twenty-four from site A2 had magnetic moments above the lower limit of the magnetometer, in spite of their increased volume. From sites B1 and B2, 2.5-cm diameter cores were drilled from the hand samples in the laboratory. The blocks were unfortunately rather thin and the cores varied between 1.5 and 2.0 cm in length. In spite of the reduced volume, these specimens were more strongly magnetized than the somewhat longer ones from sites A1 and A2 and could easily be measured on the magnetometer.

It was impossible to drill cores from the oriented blocks from site C1. The varves were so thin that the cores merely broke up into thin disks. Instead, the five blocks were cut into cubical samples of size approximately 3.5 cm. Attempts were made to spin these samples in a special holder on the magnetometer, but at the first attempt the sample shattered into fragments and could not be recovered. Thus, whereas the measurements on the other samples were made at the University College of Rhodesia, measurements on the four remaining cubical samples from site C1 were made on the sensitive astatic magnetometer at the Australian National University. The increased volume gained from having cubical samples was an advantage, since these varves were very weakly magnetized and approached the lower limit of measurement on the astatic magnetometer.

The NRM directions of the samples from sites A1 and A2 are illustrated in Figure 2. The directions are strung from the direction of the present earth's field to a direction almost reversed with respect to it. The measurements are shown with respect to the present horizontal, although at locality A the beds at both

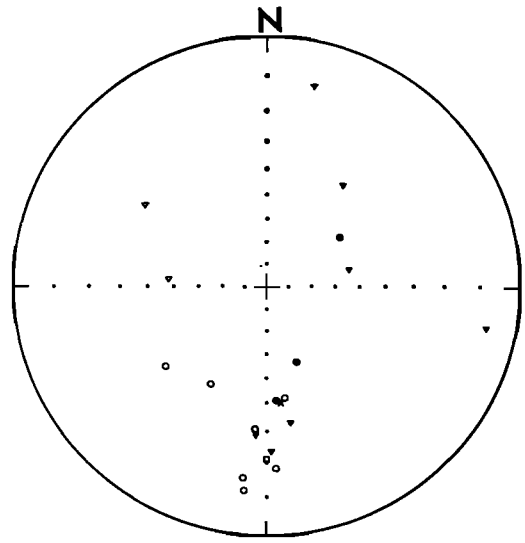


Fig. 2. NRM directions of samples from sites A1 and A2 with respect to the present horizontal. Circles, site A1; triangles, site A2. Lower hemisphere stereographic projection; solid symbols, north-seeking direction down; open symbols, south-seeking direction down. Cross, present earth's field (south-seeking down); square, dipole field (south-seeking down).

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sites dip at 32° in a direction 13° east of north. At locality B the NRM directions were similarly strung, and here the beds dip at 13° in a southeasterly direction.

Thermal Demagnetization

Two specimens from each of the sites A1, A2, B1, and B2 were thermally demagnetized by successively heating to higher temperatures and cooling in zero field. One of the two specimens from sites A1 and A2 became unmeasurable after 300°C, and the other two were still just measurable after 450° and 500°C, respectively. The thermal demagnetization curves for a specimen from each of the sites (Figure 3) show that the magnetization is composed entirely of thermally distributed components [*Irving and Opdyke*, 1965].

Specimens from sites A1 and A2 were thermally cleaned at 300°C. After this treatment only six of the ten specimens from site A1 and seven of the nine from site A2 were still measurable on the magnetometer. The directions of these specimens, however, became well grouped almost reversed with respect to the present earth's field (Figure 4). With respect to the bedding plane the samples were steeply magnetized (Table 1). The test samples from sites B1 and B2 became nearly vertically magnetized at 400°C and showed no change in direction thereafter up to 600°C. All samples from these two sites were cleaned thermally at 450°C, and after this treatment one of the thirteen samples became unmeasurable. The directions became well grouped and magnetized almost vertically downward (Figure 4, Table 1).

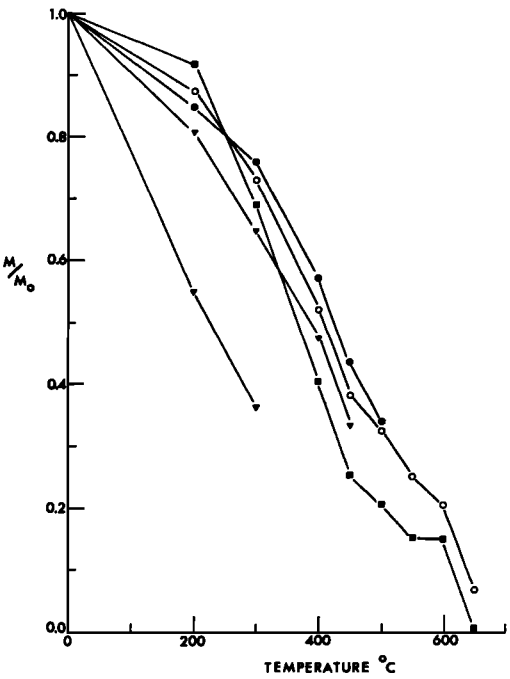


Fig. 3. Thermal demagnetization curves of a specimen from each site. Magnetic moments are normalized to the value at room temperature. Solid circles, A1; solid triangles, A2; open circles, B1; squares, B2; open triangles, C1.

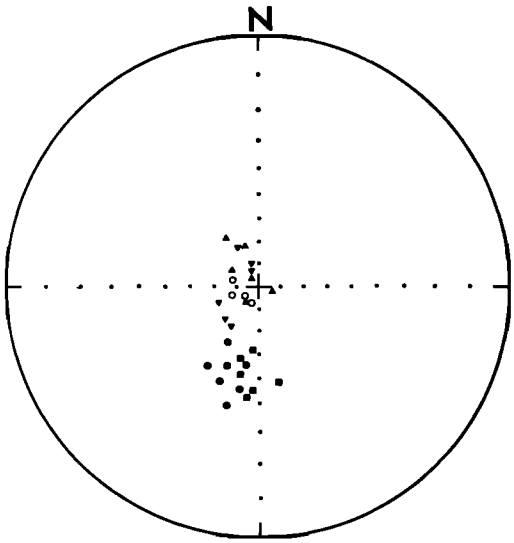


Fig. 4. Lower hemisphere stereographic projection of the directions of magnetization in the glacial varves after thermal cleaning. Directions are with respect to the present horizontal in each case, solid symbols north-seeking directions down; open symbols, south-seeking directions down. Solid circles, site A1; squares, site A2; upright triangles, site B1; inverted triangles, site B2; open circles, site C1.

The four samples from site C1 were initially magnetized approximately along the present dipole field direction. All four samples were heated to 200° and 300°C, by which stage they become steeply magnetized upward, almost reversed with respect to the other four sites (Figure 4, Table 1). Very little change in direction is observed between 200° and 300°C, and at this stage the specimens were so weakly magnetized that no further heating was attempted. *Nairn* [1960] reports NRM directions from four samples from the same site. He ob-

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TABLE 1. Site Direction for the Carboniferous of Central Africa

Site	Site Position						Mean Direction					Site Pole	
	Lat., °S	Long., °E	C	M	N	Tempera- °C	D	I	R	k	α_{95}	Lat., °S	Long., °E
A1	8.8	32.9	10	10	6	300	213½	79½	5.929	70	8	25	21
A2	8.8	32.9	24	9	7	300	165	81½	6.926	81	6½	25	37½
B1	16.4	28.2	7	7	6	450	126	86	5.876	40	10½	21	35½
B2	16.4	28.2	6	6	6	450	184	80	5.783	23	14	36	26½
C1	17.5	28.2	5	4	4	300	62½	−81	3.982	166	7	24½	11½

Notes

C, number of samples collected.
M, number of samples that could be initially measured,
N, number of samples that could be measured after thermal cleaning at the temperature indicated.
D, declination east of true north.
I, inclination measured positive downward from the horizontal.
R, resultant of N unit vectors.
k, a measure of precision [*Fisher*, 1953].
α_{95}, semi-angle of cone of 95% confidence about the mean direction.

tained both normal and reversed directions of magnetization. His normal directions (negative dips) agree very well with those obtained after thermal cleaning in this study. Nairn's two groups are not really 180 degrees apart, and this can be explained by supposing that his reversed group contained a viscous component of magnetization along the present earth's field. A viscous component added to the normal group would merely increase the divergence from 180 degrees. Nairn's dual polarities are probably real and result from fortuitous sampling of the same outcrop. His estimate of 100 years for the time for the reversal to have taken place may be an underestimate. The annual layers represented in the exposure of less than a meter in thickness cover at least 1000 years, and this seems to be sufficient time for a reversal to have occurred [*Ninkovitch et al.*, 1966].

Pole Position

Table 1 gives the parameters of the site mean directions after thermal cleaning and after restoring the beds to the horizontal in each case. In view of the geographical spread of the sampling sites, it is more appropriate to consider the grouping of the site poles than to consider the site directions, since the poles take into account the differences in site locality. The change in the positions of the site poles after restoring the beds to the horizontal is illustrated in Figure 5. The grouping of the poles is much improved. With the beds in their present attitudes, the precision of the five site poles [*Fisher*, 1953] is 7.8, and, after restoring the strata to the horizontal, this value increases to 52.4. For $N = 5$ this represents a highly significant increase in grouping at the 99% confidence level [*McElhinny*, 1964]. This corresponds to a successful application of *Graham's* [1949] fold test. In the Galula coal field (sites A1 and A2), the tilt of the beds is associated with the formation of the Rukwa rift valley [*Spence*, 1954]. Therefore, the magnetization of the beds is at least pre-Tertiary in age. The mean pole position for the Dwyka varves lies at 26½°S, 26½°E, with $A_{95} = 10½°$.

Comparison of Permo-Carboniferous Results from Africa and Australia

Irving [1966] has made a detailed study of the magnetization of rocks of Carboniferous age from Australia. Rocks of Devonian and Lower Carboniferous age from Australia have shallow magnetic inclinations, whereas rocks of Upper Carboniferous and Mesozoic age have steep inclinations. As Irving points out these results are consistent with *King's* [1962] reconstruction of Gondwanaland and with the suggestion that the south pole moved across Gondwanaland from west to east. The timing, as suggested by the paleomagnetic evidence from Australia, is different, however. King's Upper Carboniferous

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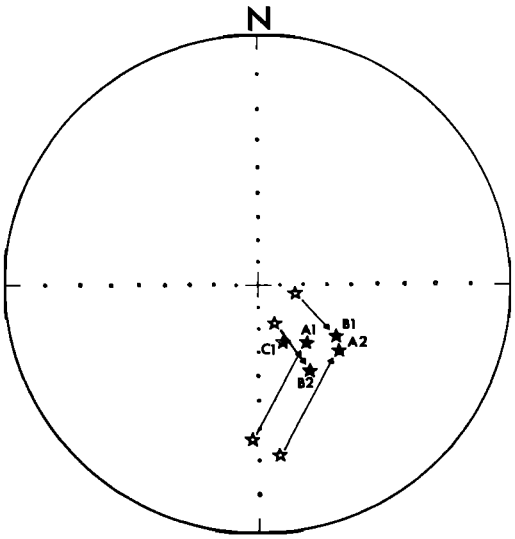


Fig. 5. Sitepoles for the Carboniferous glacial varves from Central Africa; open stars, with respect to the present horizontal; solid stars, with respect to the bedding plane. Equatorial stereographic projection.

pole is located close to the Lower Carboniferous pole deduced from the paleomagnetic data. The pole moves rapidly across Antarctica until by the Lower Permian it has the location of the Permo-Carboniferous poles from Australia (Figure 6). Irving further suggests that the Lower Kuttung (Lower Carboniferous) should be correlated with the Dwyka series of the Karroo system. As he points out, the evidence for an Upper Carboniferous age for the Dwyka is by no means firm.

The mean pole position deduced for the Dwyka series from this study coincides, within the experimental error, with King's Upper Carboniferous pole, but, using King's reconstruction (Figure 6), it also coincides with that of the Mugga Mugga porphyry (Upper Silurian or Lower Devonian [*Briden,* 1966]) and the Lower Kuttung (Lower Carboniferous) pole from Australia. King's Upper Carboniferous designation undoubtedly arises from assigning the Dwyka series to this period. As Irving points out, the paleomagnetic evidence now confirmed by this investigation places the Dwyka series more logically in the Lower Carboniferous. (Recently, E. P. Plumstead (private communication) has identified plant fossils in the Orange Free State lying directly upon Dwyka tillite, the plants include *Leptophloeum australe,* which is regarded as Upper Devonian or Lower Carboniferous. This new fossil evidence provides independent support for the conclusions drawn here.)

Irving and Parry [1963] have noted that rocks of late Carboniferous and Permian age tend to have positive polarity, and they have proposed that this period be referred to as the *Kiaman magnetic interval,* during which the earth's magnetic field had constant polarity. *Irving* [1966] notes that there is now a total of forty-one rock units, of this age with the same polarity. Confirmation of the existence of the Kiaman interval has now been provided from paleomagnetic studies of redbeds in the Rocky Mountains [*McMahon and Strangway,* 1967]. Since one of the sites (C1) in this study has negative polarity, this suggests a pre-Kiaman age for these varves, which is consistent with their suggested Lower rather than Upper Carboniferous age.

Opdyke [1964] reports some paleomagnetic measurements on the K3 beds from three of the coal fields in southern Tanzania. The K3 beds in the Ketewaka-Mchuchuma coal field and the Songwe-Kiwira coal field have paleomagnetic directions that can be shown to be significantly different and more to the east of those from the K3 beds of the Galula coal field. When Opdyke's measurements are reanalyzed in terms of two separate poles, these poles are found to lie on the polar path across Gondwanaland, as shown in Figure 6. The pole from the Songwe-Kiwira and Ketewaka-Mchuchuma K3 beds lies farther to the east and, on King's reconstruction, corresponds to the Permian and Upper Kuttung poles from Australia. The Galula K3 beds give a pole lying between the Lower Permian and Lower Carboniferous poles and very close to that for the Paterson toscanite from the Hunter valley, New South Wales (Figure 6). *Opdyke* [1964] points out that the divergence of the directions between the coal fields might be due to poor correlation between them and due to a genuine movement of the pole. Although he thought it likely that all his K3 beds were Lower Permian in age, an Upper Permian age for the Galula K3 beds could not be ruled out. The possibility that these beds might be Upper Carboniferous had not occurred at that time, and there is, like-

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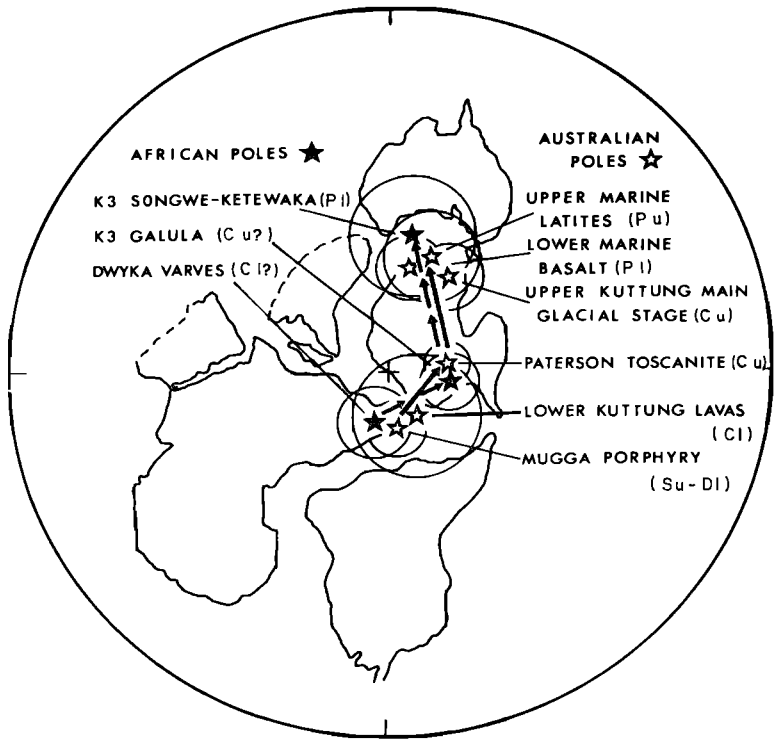


Fig. 6. Upper Paleozoic polar movement relative to Africa and Australia superimposed on *King's* [1962] reconstruction of Gondwanaland for the Upper Carboniferous. Australian poles are from *Irving* [1966] and *Irving and Parry* [1963]. African poles are those summarized in Table 2 of this paper. Circles of 95% confidence are shown for each of the poles.

wise, no reason to rule out this possibility either. A correlation with the Paterson toscanite places them in this period.

Hart [1960], on the basis of a microflora assemblage extracted from the underlying K2 beds of the Ketewaka-Mchuchuma coal field, has correlated these beds with the Permian of Australia. These K2 beds are correlated with the Greta coal measures of New South Wales and the Holmewood shale of the Irwin basin. This correlation remains consistent with the paleomagnetic evidence. The paleomagnetic evidence casts some doubt on the correlation of the K2 and K3 beds in the Ketewaka-Mchuchuma coal field with those in the Galula coal field.

Conclusions

We conclude, therefore, by proposing that the Dwyka series at the base of the Karroo system in Rhodesia, Zambia, and Tanzania be placed in the Lower Carboniferous. The base of the Karroo system in these areas is normally correlated with the Dwyka tillite of South Africa, and our results consequently suggest that the Dwyka series there is also of this age. It appears also that the K3 beds from the three coal fields of southern Tanzania are not necessarily of the same age and may range from Upper Carboniferous to Lower Permian. Table 2 summarizes the proposed ages of the various beds together with their corresponding paleomagnetic poles, derived either from this study or by recalculation from *Opdyke* [1964]. We suggest that in the future these results be taken as being representative of the Permo-Carboniferous of Africa.

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TABLE 2. Summary of Proposed African Pole Positions for the Permo-Carboniferous

Formation and Locality	Age	N	Latitude, °S	Longitude, °E	K	A_{95}	Reference
Glacial varves, Central Africa	C1?	5	26½	26½	52.4	10½	This paper
K3 redbeds, Galula coal field, Tanzania	Cu?	5	45½	40	90.9	8	Recalculated from *Opdyke* [1964]
K3 redbeds, Ketewaka-Mchuchuma and Songwe-Kiwira coal fields, Tanzania	P1	4	27	89	35.2	15½	Recalculated from *Opdyke* [1964]

Notes

N, number of sites.

K, precision of site poles [*Fisher*, 1953].

A_{95}, semi-angle of cone of 95% confidence for the mean pole position.

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